GATE-AII BRANCHES
Engineering Mathematics



Probability and Statistics

Lecture No.- 16



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Recap of Previous Lecture



Topic

questions

~~Concepts of~~ Gaussian distribution and uniform distribution.

Topics to be Covered



Topic Question based on uniform distribution and gaussian distribution

Topic Poisson distribution



Topic : Probability and Statistics

#Q. Let X_1, X_2, X_3, be three independent and identically distribution random variables with Uniform distribution on [0,1]. Find the probability $p[X_1 + X_2 \leq X_3]$



Topic : Probability and Statistics

M.W

#Q. If X is a Gaussian Distributed Random variable with Mean = 30 and Standard Deviation = 5, then find $P(|X-30| < 5)$



Topic : Probability and Statistics

#Q. Let X be zero mean unit variance Gaussian Random variable, find

$\boxed{E(|X|).}$

X is a Gaussian random variable $X \sim N(\mu, \sigma^2)$

$\mu = 0 \quad \sigma^2 = 1$

$X \simeq N(0,1)$

$E[|X|]$

$E[X] = \int_{-\infty}^{\infty} x f(x)\, dx$

$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{\frac{-(x-\mu)^2}{2\sigma^2}} = N(\mu, \sigma^2)$

Plug in $\sigma = 1$, $\mu = 0$

$= N(0,1)$

$\boxed{N(0,1) = \frac{1}{\sqrt{2\pi}} e^{\frac{-x^2}{2}}}$

EXPECTED value

$$E[|X|] = \int_{-\infty}^{\infty} |x| \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx$$

Not prob SURE

$$= 2\int_0^{\infty} x \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx$$

$$= \frac{2}{\sqrt{2\pi}} \int_0^{\infty} x e^{-\frac{x^2}{2}} dx \qquad \frac{x^2}{2} = t$$

$$= \frac{2}{\sqrt{2\pi}} \int_0^{\infty} e^{-t} dt \qquad x\,dx = dt$$

$$= \sqrt{\frac{2}{\pi}} \int_0^{\infty} e^{-t} dt$$

$$= \sqrt{\frac{2}{\pi}} \left[-e^{-t}\right]_0^{\infty} = \sqrt{\frac{2}{\pi}} \left[-e^{-\infty} + e^{0}\right] = \sqrt{\frac{2}{\pi}} = 0.8 \text{ Ans}$$

$|x|$ → even Function

$y = x^3$

$\int_{-a}^{a} f(x) = 2\int_0^{a} f(x)\,dx$ ↳ even

$\Rightarrow \int_{-\infty}^{0} -x\,dx + \int_0^{\infty} x\,dx$

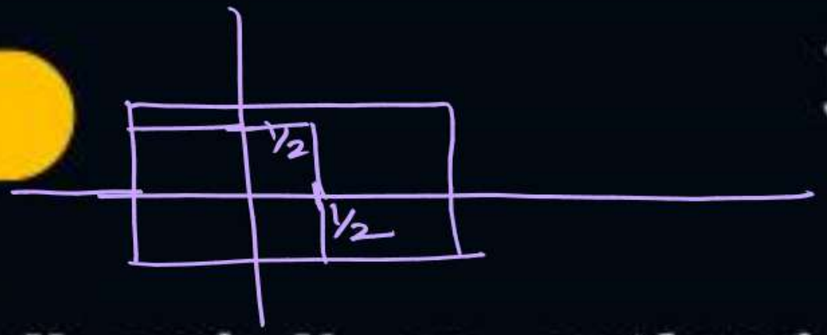
$Var(X) = \sigma^2 = 1$

$V(x) > 0$

$V(-x) = (-1)^2 Var(x)$

Using Property of definite

Topic : Probability and Statistics



#Q. Two independent random variables X and Y are uniformly distributed in the interval [−1, 1]. The probability that max[X, Y] is less than 1/2 is

A 3/4

B 9/16

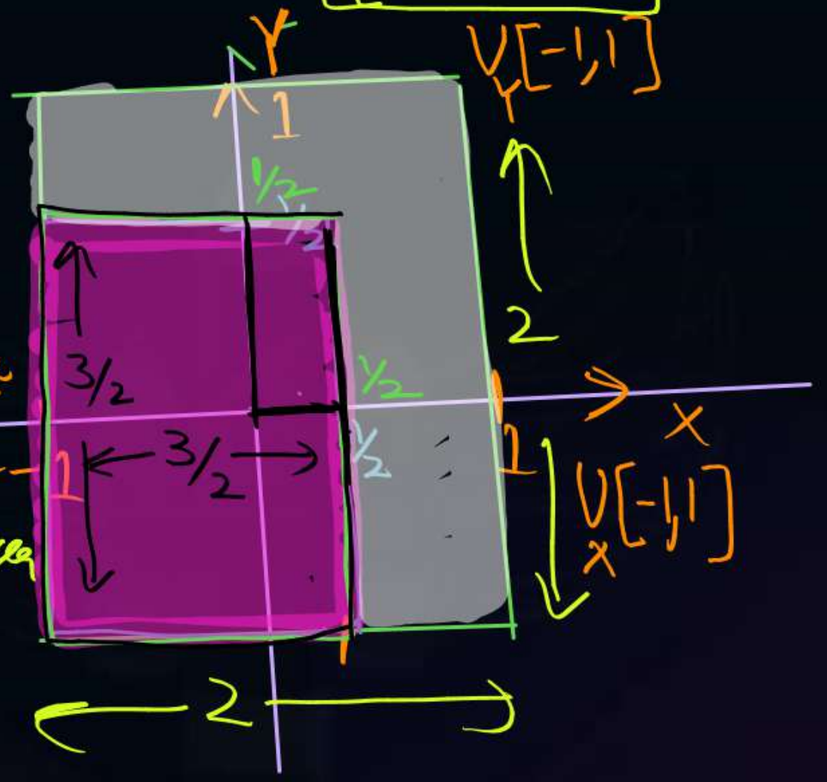
C 1/4

D 2/3

$X \sim U[-1,1]$
$Y \sim U[-1,1]$

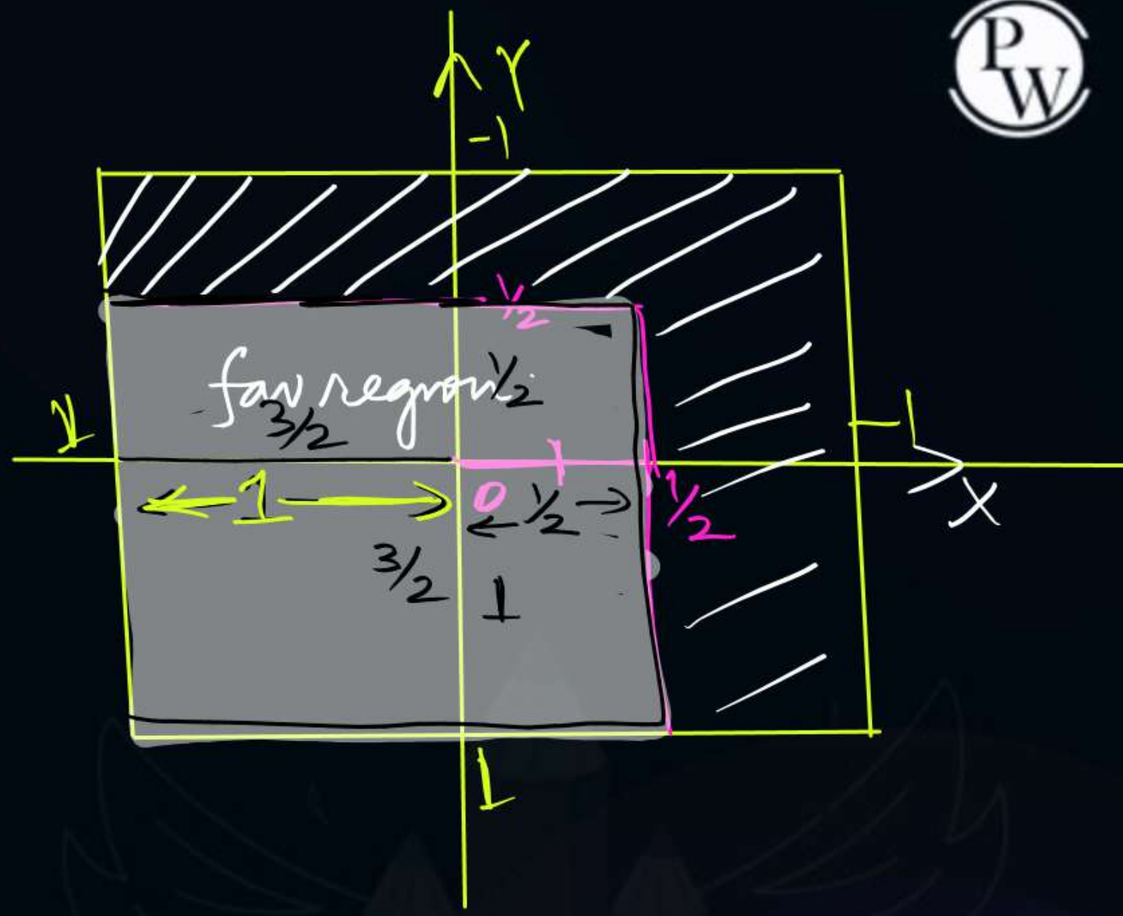
$$P\left[\max(X,Y) \le \frac{1}{2}\right] = \frac{\text{Area of Pink region}}{\text{Area of Total region}} = \frac{\text{Pink rectangle Area}}{\text{Total}}$$

$$= \frac{3/2 \times 3/2}{2 \times 2} = \frac{9}{16}$$



$$P(\max(x,y) \le \tfrac{1}{2})$$

$$\Rightarrow \frac{\frac{3}{2}\times\frac{3}{2}}{2\times 2} = \frac{9}{16}$$





Topic : Probability and Statistics

$1^2+(2)^2=5$
$=\sqrt{5}$
always Positive

#Q. The probability density function of a random variable X is $P_x(x) = e^{-x} for\ x \geq 0$ and 0 otherwise. The expected value of the function $g_x(x) = e^{3x/4}$ is 4.

$$f(x) = P_X(x) = e^{-x} \quad x \geq 0$$

$$E[g(x)] = \int_0^\infty g(x) f(x)\,dx = \int_0^\infty e^{3x/4} e^{-x}\,dx = 4$$

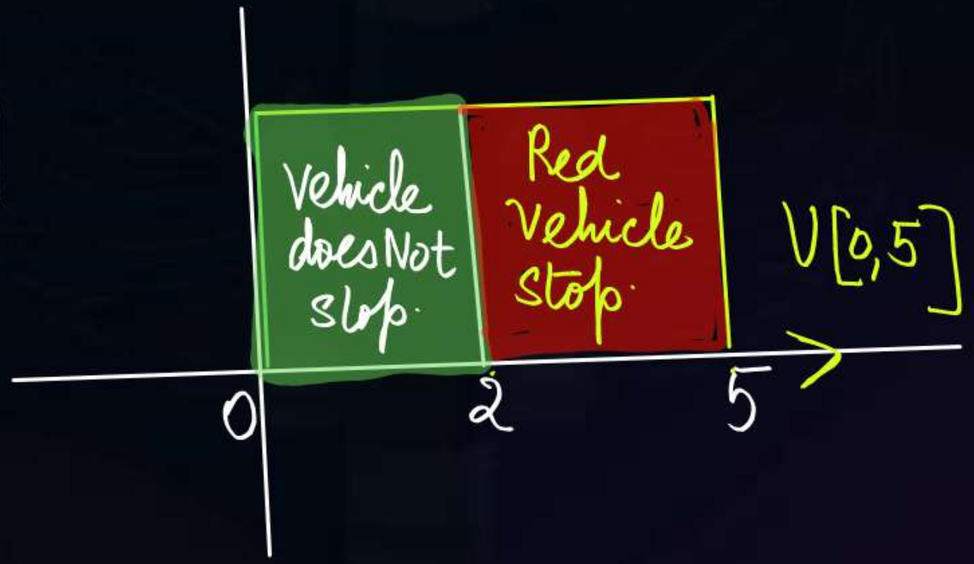


Topic : Probability and Statistics

#Q. Assume that in a traffic junction, the cycle of the traffic signal lights is 2 minutes of green (vehicle does not stop) and 3 minutes of red (vehicle stops). Consider that the arrival time of vehicles at the junction is uniformly distributed over 5 minute cycle. The expected waiting time (in minutes) for the vehicle at the junction is _____.

Expected waiting time $= E[W(x)]$

$W(x)$ = waiting time



X is Uniformly Distributed RV

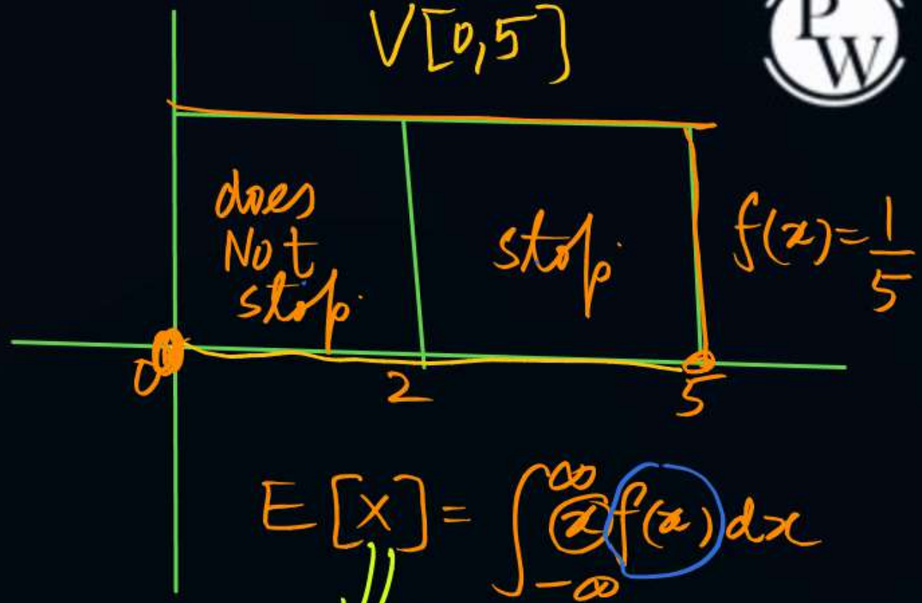
$$f(x) = \frac{1}{(b-a)} \quad U[0,5]$$

$$f(x) = \frac{1}{(5-0)} = \frac{1}{5} \quad 0<x<5$$

0 otherwise

$$f(x) = \frac{1}{5} \quad 0<x<5$$

$$W(x) = \begin{cases} 0 & 0<x<2 \\ 5-x & 2<x<5 \end{cases}$$



$$E[X] = \int_{-\infty}^{\infty} x f(x)\,dx$$

$$E[W(x)] = \int_{-\infty}^{\infty} W(x) f(x)\,dx$$

$X = W(x)$ Waiting Time



$E[W(x)]$ = Expected waiting Time

$$E[X] = \int_{-\infty}^{\infty} x f(x)\,dx$$

$$E[W(x)] = \int_{-\infty}^{\infty} W(x) f(x)\,dx$$

$$= \underbrace{\int_{0}^{2} 0 \cdot \frac{1}{5}\,dx}_{0} + \int_{2}^{5} (5-x) \cdot \frac{1}{5}\,dx$$

$$= \frac{1}{5}\int_{2}^{5} (5-x)\,dx \Rightarrow \frac{1}{5}\left[\left(5x - \frac{x^2}{2}\right)\right]_{2}^{5} = 0.9 \text{ min} = \underline{54 \text{ SECOND}}$$

expected waiting Time

$$f(x) = \frac{1}{5} \quad 0 < x < 5$$

$$W(x) = \begin{cases} 0 & 0 \le x \le 2 \\ 5-x & 2 \le x \le 5 \end{cases}$$

✓ ✓ ✓ X, 0 Concise



Topic : Probability and Statistics

CS
4/5

#Q. Suppose Y is distributed uniformly in the open interval (1, 6). The probability that the polynomial $3x^2 + 6xy + 3y + 6$ has only real roots is (rounded off to 1 decimal place)

$(\vec{1}, 6)$

$3x^2+6xy+3y+6$ Has only Real Roots

$a=3 \quad b=6y \quad c=3y+6$

If equation Represent The real Root $ax^2+bx+c=0$ quadratic equ^n

$b^2-4ac \geq 0$

$36y^2-4\times3\times(3y+6)\geq 0$

$= 36y^2-12(3y+6)\geq 0$ ✓

$= 36y^2-36y-72\geq 0$

$y\geq 2$

$y^2-y-2\geq 0$

$y^2-2y+y-2\geq 0$

$y(y-2)+1(y-2)\geq 0$

$(y+1)\geq 0$

$y\geq -1$ ✗

$Y \geq 2$, $y \geq -1$

X

$$f(x) = \begin{cases} \frac{1}{(6-1)} & 1 < x < 6 \\ 0 & \text{otherwise} \end{cases}$$

$$f(x) = \begin{cases} \frac{1}{5} & 1 < x < 6 \\ 0 & \text{otherwise} \end{cases}$$

$$P(Y \geq 2) = \int_2^6 \frac{1}{5}\, dx = \frac{4}{5} \quad \text{Ans}$$

$V[1, 6]$

Read the question



Properties of Guassian random variable:

If 'X' and 'Y' Two Guassian Distributed random variable

If X, Y both $N(\mu, \sigma^2)$

$X \sim N(\mu_X, \sigma_X^2)$



✓ Then sum of random variable

$Z = X + Y$

Z is also guassian Distributed

✓ $Z = X - Y$ = difference of guassian random var.

$Y \sim N(\mu_Y, \sigma_Y^2)$



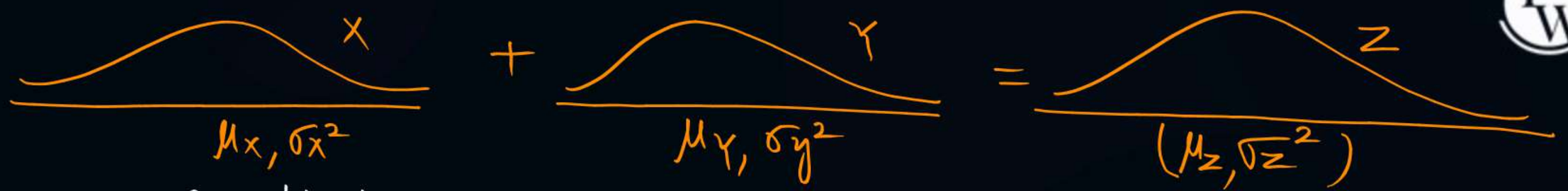
$N(\mu_Z, \sigma_Z^2)$

$Z = X + Y$



$Z = X - Y$

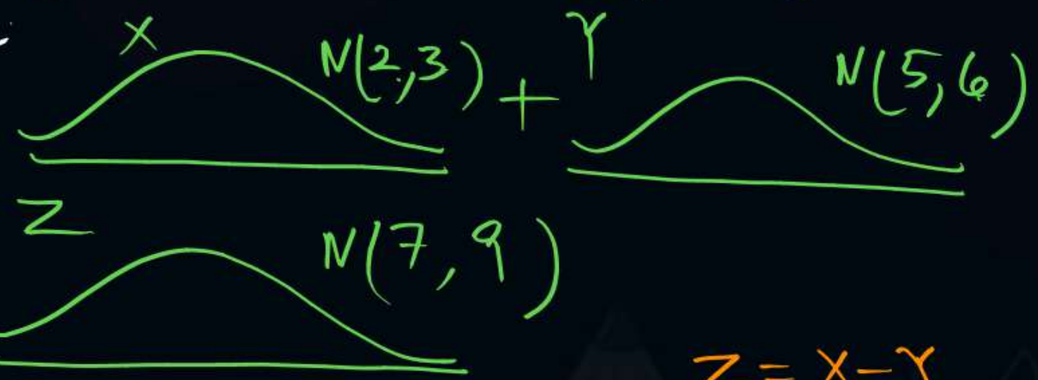


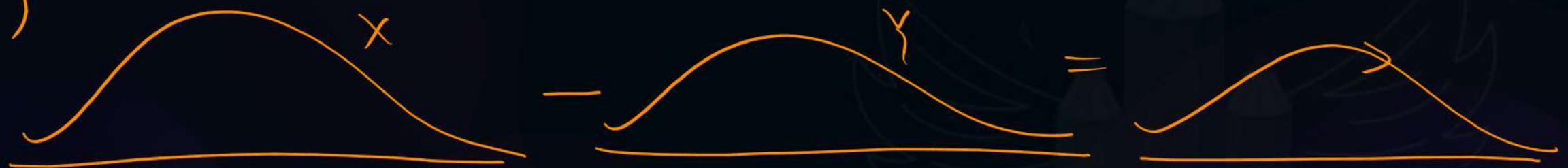
1) 

Combined mean or variance

$Z = X + Y$

$$\mu_Z = \mu_X + \mu_Y$$
$$\sigma_Z^2 = \sigma_X^2 + \sigma_Y^2$$

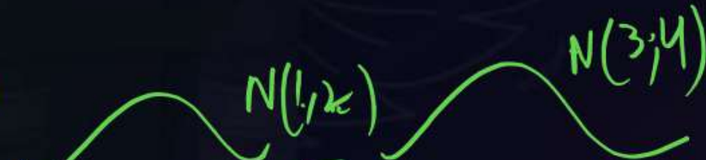


2) 

$Z = X - Y$

$$\mu_Z = \mu_X - \mu_Y$$
$$\sigma_Z^2 = \sigma_X^2 + \sigma_Y^2$$

→ var can't be negative



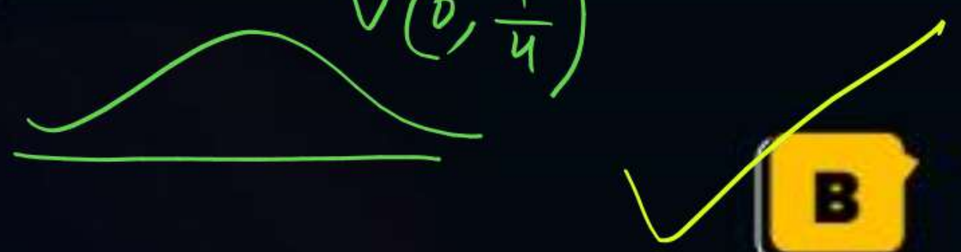
$= N(-2, 6)$

Topic : Probability and Statistics

$V(ax) = a^2 V(x)$

$V(ax+b) = a^2 V(x)$

#Q. Let U and V be two indepdent zero mean Gaussian random variables of variances ¼ and 1/9 respectively. The probaility P{3V≥ 2U}is-



A 4/9

B 1/2

C 2/3

D 5/9



$P(3V \geq 2V) = P(3V - 2V \geq 0)$

$X = 3V - 2V$

$\mu_X = 3\mu_V - 2\mu_U$

$= 3 \times 0 - 2 \times 0 = 0$

$\sigma_X^2 = (3)^2 \sigma_V^2 + (2)^2 \sigma_U^2$

$= 9 \times \frac{1}{9} + 4 \times \frac{1}{4} = 2$

$P(X \geq 0)$

$P\left(\frac{X-\mu}{\sigma} \geq \frac{0-\mu}{\sigma}\right) = P\left(Z \geq \frac{0-0}{\sqrt{2}}\right) = P(Z \geq 0) = \frac{1}{2}$



Topic : Probability and Statistics

#Q. Let x_1, x_2 be two idenpendent normal random variable with means μ_1, μ_2 and standard deviation σ_1 , σ_1 respectively. Consider $Y = X_1 - X_2$; $\mu_1 = \mu_2 = 1$, σ_1 , $\sigma_1 = 2$

$\mu_1 = 1$ $\quad [\sigma_1 = 1$

$\mu_2 = 1$ $\quad \sigma_2 = 2]$ $\qquad Y = X_1 - X_2$

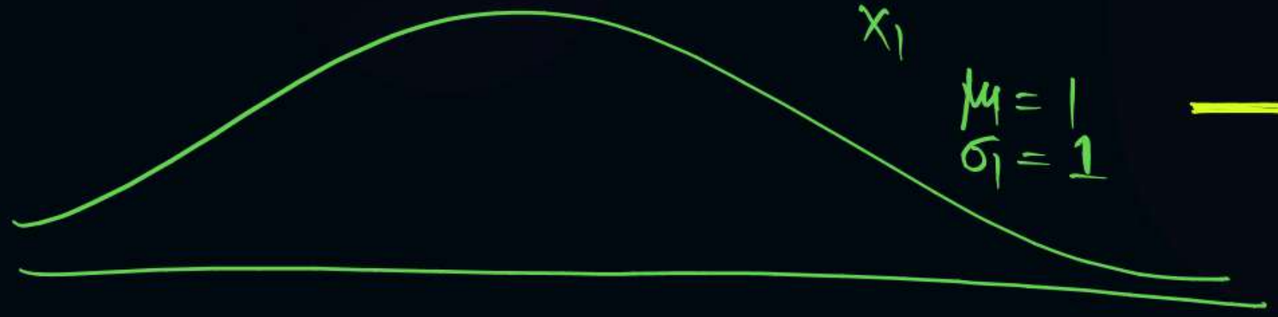
A Y is normally distribution with mean 0 and variance 1

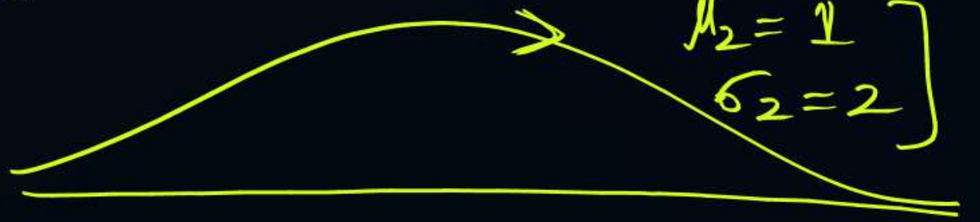
B Y is normally distribution with mean 0 and variance 5

C Y has mean 0 and variance 5, but is NOT normally distribution

D Y has mean 0 and variance 1, but is NOT normally distribution

2011





$Y = X_1 - X_2$

Y is also guassian distributed

$\mu_{Y_1} = \mu_{X_1} - \mu_{X_2} = 1 - 1 = 0$

$\sigma_{Y_1}^2 = \sigma_{X_1}^2 + \sigma_{X_2}^2$

$\sigma_{Y_1}^2 = (1)^2 + (2)^2$

$= 5$

$Y = X_1 - X_2$



Y — mean = 0

variance = 5

Standard deviation = $\sqrt{\text{variance}} = \sqrt{5}$

$N(0,5)$

TUF { DPP01, DPP02 }

DPP03

Rahul Sir PW

THANK - YOU